

SDACS MATLAB Room Mapping Guide

Purpose

This guide explains how to start the MATLAB room map without needing the full troubleshooting history. It also points to simplified CSV files that keep only the columns needed for plotting.

1. What changed from the original plan

Originally, the plan was to estimate node locations using RSSI and use SPL as the z-coordinate. That is still possible later, but it should not be the first MATLAB version.

Original idea	Current recommendation	Why
x/y from RSSI	x/y from measured room coordinates	RSSI is useful for rough proximity, but too unstable for accurate room position.
z from SPL	z from band ratios or tone ratios	SPL is not calibrated yet. Band ratios are cleaner and already working.
one spike value	choose a metric by test	Different tests need different z-values: 1 kHz, bass, mid, presence, or high.

Key message

Build the plot with band ratios first. Treat `db_spl` as an uncalibrated reference value until SPL calibration is proven.

2. Is the data clean enough for MATLAB?

Yes. The current telemetry is clean enough to build and test the MATLAB import and plotting workflow. The file has one header, useful numeric columns, and the new room-analysis fields.

Important limitation: this test data is still from one node, node04. That is enough to build the MATLAB code structure, but a real room map needs synchronized captures from node01, node02, node03, and node04.

Test label	Rows	Mean sample rate	Good SR rows	Band sum	Mean 1k ratio	Mean contrast dB	1k rows
quiet room	17	47,975.708	17	1	0.071	-5.809	1
tone 1khz 100db	19	48,034.356	19	1	0.131	-2.657	8
low frequency test	19	48,082.861	19	1	0.06	-6.356	1

The band ratio sum is close to 1.0, which means the frequency-band normalization is working. The sample rate is close to 48 kHz, which means the capture timing is healthy enough for feature-based plotting.

3. Files included for Amy

File	Use
<code>sdacs_matlab_relevant_telemetry.csv</code>	Simplified per-row telemetry. Use this first in MATLAB.
<code>sdacs_capture_summary_for_matlab.csv</code>	Per-capture summary. Useful for checking the test results quickly.
<code>sdacs_node_positions_template.csv</code>	Template where Amy can enter the actual room x/y positions of each node.

4. Columns Amy should care about

Column or group	Meaning
node_id, request_id, test_label	Identify which node, capture, and test condition each row belongs to.
effective_sample_rate_hz, sample_rate_ok	Quality check. Prefer rows where sample_rate_ok is true.
tone_1khz_ratio	Best z-value for a 1 kHz spike map.
tone_1khz_local_ratio, tone_1khz_contrast_db	Extra diagnostics for comparing 1 kHz against nearby frequencies.
band_bass_ratio	Best z-value for low-frequency buildup.
band_mid_ratio, band_presence_ratio	Useful for speech and presence-region energy.
band_high_ratio	Useful for high-frequency energy or reflection tests.
dominant_band_name, dominant_band_ratio	Quick view of which band is strongest in a row.
db_spl	Uncalibrated reference only. Do not use as final SPL yet.

5. Recommended graph approach (You already get this... but for the sake of documetation)

Start with a simple 3D scatter plot:

x = node x-position in the room

y = node y-position in the room

z = one selected acoustic metric, such as tone_1khz_ratio or band_bass_ratio

color = same z-value or dominant band

label = node_id

MATLAB graph	z-value
1 kHz spike map	mean(tone_1khz_ratio)
Bass buildup map	mean(band_bass_ratio)
Speech/mid map	mean(band_mid_ratio)
Presence map	mean(band_presence_ratio)
High-frequency map	mean(band_high_ratio)
Uncalibrated SPL map	mean(db_spl), clearly labelled as uncalibrated

6. RSSI and SPL guidance

RSSI can still be used, but it should not be the main way to place nodes on the graph. RSSI changes with walls, equipment, people, antenna orientation, and reflections. It is better for rough proximity than precise coordinates.

SPL can still be included later, but only after calibration is validated. For now, MATLAB should use the frequency-band ratios as the main graph height.

7. Possible MATLAB skeleton code

This code is intentionally a skeleton. It gives the structure and leaves room to decide which graph and metric to focus on **(feel free to completely ignore this or modify it as you see fit as this is just an example I found)**.

```
%% SDACS MATLAB Room Mapping Skeleton

% Goal:

% Explore SDACS acoustic telemetry and begin building a room mapping workflow.

%

% General idea:

% - Import telemetry data

% - Import node position data

% - Choose an acoustic feature to investigate

% - Compare nodes by location

% - Visualize frequency-spike strength in the room

%

% Note:

% The current dataset may only contain one node.

% That is enough to test importing and plotting logic,

% but a real room map requires multiple nodes.


clear; clc; close all;


%% 1. Load data

% TODO: Confirm these filenames match the files in your MATLAB folder.

telemetryFile = "sdacs_matlab_relevant_telemetry.csv";

positionFile  = "sdacs_node_positions_template.csv";


T = readtable(telemetryFile, "TextType", "string");

P = readtable(positionFile,  "TextType", "string");


%% 2. Inspect the data

% TODO:

% Look at the imported tables and identify which columns are useful.

% Useful commands:

%   head(T)

%   summary(T)
```

```

% T.Properties.VariableNames

disp("Telemetry columns:");
disp(T.Properties.VariableNames');

disp("Position columns:");
disp(P.Properties.VariableNames');

%% 3. Clean or filter the telemetry

% TODO:

% Decide what rows should be kept before plotting.

%

% Suggested checks:

% - Is sample_rate_ok true?

% - Is the test_label the one you want?

% - Are the acoustic feature columns numeric?

% - Are there missing values?

% Example placeholder:

% T = T(... , :);

%% 4. Choose what acoustic feature to investigate

% TODO:

% Pick one test condition and one z-axis feature.

%

% Possible test labels:

% - quiet_room

% - tone_1khz_100db

% - low_frequency_test

%

% Possible z-axis metrics:

% - tone_1khz_ratio

% - band_bass_ratio

% - band_low_mid_ratio

% - band_mid_ratio

```

```

% - band_presence_ratio
% - band_high_ratio
% - dominant_band_ratio
% - db_spl    % uncalibrated estimate only

targetLabel = "tone_1khz_100db";
zMetric = "tone_1khz_ratio";

%% 5. Prepare the selected dataset

% TODO:
% Filter the telemetry table to the selected test label.
% Then make sure the chosen zMetric column is numeric.

% Example placeholder:
% A = T(...);

%% 6. Summarize the data by node

% TODO:
% Each node will have multiple telemetry rows during one capture.
% Decide how to summarize those rows.
%
% Possible summary values:
% - mean
% - max
% - median
% - number of detections
%
% Useful MATLAB functions:
% - findgroups()
% - splitapply()
% - groupsummary()

% Example placeholder:
% S = ...

```

```

%% 7. Combine telemetry with node positions

% TODO:

% Join the summarized telemetry table with the node position table.

%

% The position table should include at least:

% - node_id

% - x_m

% - y_m

%

% Optional:

% - height_m

% - rssi_dbm

%

% Useful MATLAB function:

% - innerjoin()

% Example placeholder:

% M = ...

%% 8. Create a basic 3D plot

% TODO:

% Plot node position against acoustic feature strength.

%

% Suggested mapping:

% - x-axis: node x position

% - y-axis: node y position

% - z-axis: selected acoustic feature

%

% Useful MATLAB function:

% - scatter3()

figure;

% TODO: Add scatter3() here

```

```

grid on;

xlabel("Room X position (m)");
ylabel("Room Y position (m)");
zlabel(zMetric);

title("SDACS Room Frequency Spike Map");

colorbar;

%% 9. Add node labels

% TODO:

% Add text labels so each point shows the node ID.

%

% Useful MATLAB function:

% - text()

%% 10. Optional: try surface interpolation

% TODO:

% Only attempt this once multiple nodes are available.

%

% With four nodes, the surface is only an estimated trend,

% not a precise acoustic model.

%

% Useful MATLAB functions:

% - meshgrid()

% - griddata()

% - surf()

%% 11. Optional: repeat with different acoustic features

% TODO:

% Try changing zMetric to compare different types of room behaviour.

%

% Examples:

% zMetric = "tone_1khz_ratio";

% zMetric = "band_bass_ratio";

% zMetric = "band_mid_ratio";

```

```

% zMetric = "band_presence_ratio";

% zMetric = "band_high_ratio";

%% 12. Notes for interpretation

% TODO:

% Think about what the graph is actually showing.

%

% Important:

% - Band ratios are currently more reliable than calibrated SPL.

% - db_spl should be treated as an uncalibrated estimate for now.

% - RSSI may be useful as supporting information, but not as a precise

%   room coordinate system.

% - A full room map requires synchronized captures from multiple nodes.

```

8. node_positions.csv template explained

Replace these coordinates with actual measured locations.

```

node_id,x_m,y_m,height_m,notes
node01,0,0,1.2,replace with actual position
node02,4,0,1.2,replace with actual position
node03,0,3,1.2,replace with actual position
node04,4,3,1.2,replace with actual position

```

9. Bottom line

The data is clean enough to start the MATLAB pipeline now. The final room map will require four-node synchronized data, but the import, grouping, and 3D plotting structure can be built with the simplified CSV provided here.